

Efficient Audiovisual Speech Processing via MUTUD: Multimodal Training and Unimodal Deployment



The hypothesis is clear, specific, and addresses a known issue in zero-shot semantic segmentation (ZS3), namely the seen-bias problem, by proposing a Swin Transformer backbone and a pixel-text score map loss. **The experimental design is highly rigorous and directly tests the hypothesis.** Experiments 1 and 2 evaluate the proposed method on two standard benchmarks (PASCAL VOC and PASCAL Context) across multiple seen/unseen splits, which demonstrates robustness. The chosen baselines (DeViSE, SPNet, ZS3Net, CSRL, JoEm) are appropriate and representative of the field. The evaluation metrics (mIoUs, mIoUu, and their harmonic mean hIoU) are the gold standard for generalized zero-shot learning, ensuring that improvements on unseen classes do not come at the catastrophic expense of seen classes. Finally, Experiment 3 provides a well-structured ablation study that isolates the individual contributions of the Swin Transformer backbone (compared to a CNN baseline) and the proposed auxiliary loss, directly validating the two core claims of the hypothesis. **Overall, the methodology is sound, controlled, and comprehensive.**